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Summarized Description of Data Loss Protection Project:

- Designed a **comprehensive Insider Threat Program** to mitigate **data leaks and unauthorized disclosures** related to Oscorp's controversial new medication.
- Implemented **Microsoft Data Loss Prevention (DLP) policies** across **Office 365, Azure, SharePoint, and endpoint devices**, preventing unauthorized transfers of sensitive files.
- Deployed **automated alerting and monitoring** for **intellectual property, personally identifiable information (PII), and financial data**, ensuring real-time response to potential breaches.
- Enforced **USB port restrictions**, preventing unauthorized copying of sensitive data to external storage devices.
- Conducted **data discovery and classification** to tag and label highly sensitive information, applying **Azure Information Protection (AIP) policies** for controlled access.
- Introduced **physical security audits**, ensuring that confidential printed materials were securely stored to mitigate **photo-based exfiltration risks**.

Data Loss Protection

By Shavaughn Hamilton

An emergency executives meeting at Oscorp was held after reports of serious side effects of Oscorp's new medication hit the press.

Norman Osborn and Harry Osborn are extremely concerned. They suspect an employee has leaked the information to the press.

Oscorp use Microsoft Office 365 and The Microsoft Azure cloud. Internally, they use Microsoft SharePoint.

Problem: Oscorp wants you to design a comprehensive "Insider Threat" program to monitor and stop employees from stealing and leaking sensitive data outside the organisation.

Practical Assessment Solution: DLP

They should implement a Data Loss Protection (DLP) that scans the environment to see if any sensitive information is leaving the organization and generates an alert for a cybersecurity professional/ investigator. They should be tactical with their implementation by classifying the data they have and then labeling them in Microsoft Office 365, Microsoft Azure cloud, Microsoft SharePoint, and desktops. Certain keywords, and information such as credit cards, personal identifiable information, and intellectual property and the size of files being transferred, whether through USB or Email should be blocked or trigger an alert and be actively monitored. If there are many false positive alerts daily, more cybersecurity professionals monitoring would be recommended or the alert triggers need to be optimized. Also, The principle of least privilege and separation of duties should also be utilized.

The first step in establishing an effective Insider Threat program is to ensure that identity and access management (IAM) is managed properly, which we covered in practical assessment of the previous module.

Therefore, our focus here will be on DLP and data security measurements.

The first step, which is a low-hanging fruit would be to block access to any USB flash drives. Simply stopping employees from copying data to external USB flash drives and hard drives is a simple yet effective control measures that's technically free to implement.

The next step would be to undertake a data discovery activity to ensure that the cyber security team have visibility over all data, specially sensitive data. Since this is an urgent undertaking, the focus should be on data related to Oscorp's controversial drug. We know from previous assessments that the formula is stored on a Microsoft SQL database so that would be our first location.

Next, a search through Microsoft SharePoint to discover any files related to the drug.

Finally, tag and label all data related to the drug as Highly Sensitive using Microsoft Azure AIP.

Since Oscorp uses Microsoft products, then a DLP solution from Microsoft is probably the most cost effective and easiest to implement. If you do a research, you will find that this product from Microsoft can do the trick.

Configure Strict DLP rules to prevent sending or copying any files tagged as "Highly Sensitive".

Finally, conduct a physical search on desks to try and find any physical files related to the drug. They should all be kept in a secure file cabinet to reduce the risk of employees taking photos of sensitive files using their phones.